

HEALTHCONNECT EXPERIENCE LAB

WEEK 6 PROJECT SUMMARY

Integration, Advanced Analytics, Validation & Decision Support

Progression — WEEK 5 OUTPUT → IMPROVE → INTEGRATE → VALIDATE → DOCUMENT → PREPARE WEEK 7

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1. Executive Summary

Week 6 advanced the HealthConnect work from descriptive and diagnostic analysis toward a more robust, integrated and testable decision-support framework. The priority Week 5 findings were revalidated, deepened through robustness testing and assessed jointly. Booking Lead Time remains the primary analytical signal, while Recorded Previous No-Shows provides complementary differentiation. Integration with Data Science independently reinforced selected findings, nuanced the interpretation of Distance to Clinic and Age, and generated additional validation questions without converting predictive relevance into causal evidence.

Week 6 Outcome — HealthConnect now has an analytically validated prioritisation framework for testing targeted attendance-support interventions, supported by validated KPIs, explicit interpretation guardrails and a measurement approach that prepares the project for Week 7.

2. Planned Work & Work Completed

Week 6 Objective	Work Completed
Deepen Week 5 findings	Robustness testing of Booking Lead Time and Recorded Previous No-Shows
Investigate combined relationships	Lead Time × Previous No-Shows matrix and additive vs interaction model comparison
Validate KPIs	Revalidated No-Show Rate and Reminder Coverage; diagnostic metrics reclassified
Identify highest-impact findings	Evidence hierarchy: Primary / Supporting / Secondary / No Standalone Prioritisation
Translate findings into actions	Specific, measurable recommendations framed as intervention tests
Collaborate across tracks	Analytics → Data Science handoff followed by Integration Back
Improve dashboard/report	Targeted updates to MONITOR, DIAGNOSE and PRIORITIZE
Prepare Week 7	Testing requirements, measurement logic and guardrails defined

3. Key Improvements from Week 5

Week 5	Week 6 Improvement
Booking Lead Time identified as primary signal	Robustness supported through alternative segmentation, statistical testing, modelling and temporal stability
Previous No-Shows used as supporting signal	Sparse categories regrouped and signal validated as complementary
Analysis mainly univariate / bivariate	Joint Lead Time + Previous No-Shows evaluation
Reminder Status treated within the KPI framework	Reclassified as a diagnostic comparative metric; Reminder Coverage retained as a process KPI
Distance treated as a weak contextual signal	Interpretation nuanced after Data Science review; retained for multivariable evaluation
General recommendations	Recommendations structured by action, target population, owner, KPI, baseline, success criterion and scale/stop rule
Initial analytical dashboard	Dashboard strengthened around MONITOR → DIAGNOSE → PRIORITIZE

4. Validated Analytical Findings

Finding	Key Evidence	Week 6 Conclusion
Booking Lead Time	NSR 39.13% for 0–30 days vs 63.95% for 31–60 days; +24.82 pp gap; RR ≈ 1.63	Strengthened and robust primary signal
Recorded Previous No-Shows	NSR 46.30% for 0 vs 57.83% for ≥1; +11.53 pp gap	Confirmed complementary signal
Lead Time × Previous NS	34.52% → 45.62% → 59.07% → 70.52% across combined profiles; complex statistical interaction not supported	Complementary signals; no complex deterministic score justified
Reminder Status	54.63% with no recorded reminder vs 49.86% with reminder; +4.78 pp gap	Limited differentiation; intervention-testing candidate
Distance to Clinic	Weak standalone Analytics association but predictive usefulness identified by Data Science	Contextual signal retained for multivariable evaluation

Interpretation Guardrail — These are observational findings. They support prioritisation and intervention design for testing; they do not establish causal mechanisms.

5. Validated KPI & Measurement Framework

Measure	Week 6 Status	Baseline / Value
No-Show Rate	Primary Outcome KPI	51.15% across 4,737 Attendance-Observable Appointments
Reminder Coverage Rate	Secondary Process KPI	72.68% across all 5,000 appointments
NSR by Reminder Status	Diagnostic Comparative Metric	54.63% No vs 49.86% Yes
No-Reminder Gap	Supporting Diagnostic Metric	+4.78 pp
NSR by Booking Lead Time	Diagnostic / Prioritisation Metric	By lead-time profile
NSR by Previous No-Shows	Diagnostic / Refining Metric	By recorded-history profile

KPI Principle — A measure is not promoted to KPI status simply because it can be calculated. A KPI must be linked to a defined decision, action or management process.

6. Cross-Track Integration — Data Analytics ↔ Data Science

Cross-track collaboration was conducted with AYDEN NGNINTEDEM DEMANOU — Data Science Intern, HealthConnect Pod 01. Data Analytics provided the validated outcome and analytical-population definitions, KPI logic, prioritised findings, data-quality limitations and interpretation guardrails. Data Science returned modelling evidence and feature-level insights, which were then reviewed and integrated back into the Analytics decision-support framework.

Exchange	Content / Impact
Analytics → Data Science	Analytical Evidence Brief, KPI Reference, population/outcome definitions, Lead Time / Previous NS / Reminder evidence, limitations
Data Science → Analytics	Retained and derived modelling features, including booking_lead_days_sq, previous_no_show_rate and long_lead_followup; predictive-relevance evidence
Integration Back	Booking Lead Time strengthened; Previous No-Shows confirmed; Distance and Age interpretations nuanced; Reminder retained as an

	intervention-testing candidate
Documentation change	Obsolete fixed Random Forest importance percentages were not retained because relative importance changes with the feature space
New analytical question	Distance × Reminder hypothesis identified for targeted analytical validation rather than accepted as an established mechanism

Evidence of Integration — The collaboration produced documented analytical change: validation, nuance and a new hypothesis for testing. It therefore goes beyond information exchange.

7. Decisions & Recommendations

Priority	Decision	Measurable Action
1 — PRIORITISE	Use Booking Lead Time as the primary prioritisation signal	Test targeted attendance support among longer-lead appointments rather than implementing an immediate system-wide change
2 — REFINE	Use Recorded Previous No-Shows as a complementary signal	Refine the tested population, subject to the recorded-history data-quality guardrail
3 — TEST & MEASURE	Evaluate reminder / confirmation / attendance-support approaches	Compare intervention and appropriate comparison groups over a defined window using a pre-specified success criterion

Deployment Condition — Extend, adjust or stop the intervention based on measured results. Prioritisation supports testing; it does not label an individual patient as a certain future no-show.

8. Dashboard Improvements

- **MONITOR** — validated headline KPIs, Attendance-Observable Appointments and cautious framing of the Reminder Status association.
- **DIAGNOSE** — strengthened Booking Lead Time evidence; Previous No-Shows regrouped to 0 / 1 / 2 / 3+; updated evidence hierarchy.
- **PRIORITIZE** — 2 × 2 Lead Time × Previous No-Shows matrix, decision-relevant gaps and a simplified Priority Action Framework.

Visual Principle — Only evidence-justified changes were introduced. Readability and decision value remain more important than visual density.

9. Challenges, Limitations & Methodological Decisions

Issue	Decision / Treatment
Patient-level inconsistencies	Maintain appointment record as the unit of analysis; do not reconstruct longitudinal patient histories
Missing Distance / Waiting Time	No arbitrary imputation; analyse available cases
Association vs causality	Frame findings as signals and recommendations as tests
Data Science feature importance	Interpret as predictive relevance, not causal importance
Random Forest percentages	Do not freeze obsolete percentages after derived features alter the feature space
No detailed financial / capacity data	Do not claim unsupported ROI or financial impact

Success criteria	Predefine with HealthConnect before intervention when the dataset provides no official target
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10. Week 6 Contribution to HealthConnect

The Week 6 Data Analytics contribution transforms the Week 5 findings into a more rigorous decision-support system. KPIs are stabilised, signals are ranked by robustness and decision value, Data Science integration is explicitly documented, and recommendations are now connected to target populations, owners, monitoring KPIs and testing conditions. This progression reduces the risk of analytical overstatement while making the evidence more directly usable for HealthConnect intervention design.

11. Remaining Issues & Week 7 Focus

- Test the stability of the proposed prioritisation across the selected populations and evaluation windows.
- Evaluate the measured effect of a targeted intervention relative to an appropriate comparison group.
- Validate whether recorded-history variables are sufficiently reliable for operational use.
- Determine whether additional Data Science features materially improve decision quality enough to justify operational use.
- Test stakeholder comprehension and usability of the dashboard and Priority Action Framework.
- Maintain a clear distinction between predictive relevance, analytical association and causal evidence.

Week 7 Transition — Week 6 established and integrated the decision-support framework. Week 7 should test, refine and validate end-to-end whether the proposed attendance-support intervention produces measurable improvement before wider deployment.